mdstats 0.20.180a0

PERF-P1 shared exact selection and progressive coverage

mdstats project

2026-08-15

1 Release decision

`mdstats 0.20.180a0` qualifies bounded `PERF-P1`. The release consolidates exact selection/coverage execution and removes quadratic persistent DATA7 neighbor state without changing scientific content. `PERF-P2` is next.

2 Implemented

- Add `ExactFPSState` with reusable row norms, selected state, lexical UID rank, minimum squared distances, and exact continuation.
- Hand the TARGET-DATA2C quota state directly into FPS continuation.
- Bound centroid novelty temporaries while preserving the historical subtraction/reduction numerical path.
- Preallocate the fused TARGET-DATA2C FP64 selector matrix and fill family column slices directly.
- Add progressive nested TARGET-DATA2B coverage scoring with one family loaded and scaled at a time.
- Route cKDTree query workers through the campaign coverage CPU-budget resolver [2].
- Replace DATA7's persistent dense KxK selected-distance matrix with one O(K) nearest-neighbor-minimum vector plus bounded exact pair blocks.
- Add independent legacy-oracle tests and matched CPU/memory benchmark evidence.

3 Exactness

The complete PERF-P0 native reference contains 37,633 target frames and eight coverage families. PERF-P1 preserves:

- the full 37,633x50 selector matrix byte-for-byte;
- exact FPS order through K=1024;
- all coverage report digests at 128/256/512/1024;
- worker-count invariant coverage digests; and
- regression TARGET-DATA2C and DATA7 content digests.

Scientific benchmark digest:

`ff08ca4aee884f1aaf4bf1969454bb75fc9e875eb8c29d57c46fe0100dadb12e`.

4 CPU and memory evidence

Case	Legacy	PERF-P1	Result
Full-reference FPS K=1024	1.550 s	0.657 s	**57.61% faster**
Wide FPS 4000x128, K=512	0.182 s	0.049 s	**73.25% faster**
DATA7 K=8192 wall	1.954 s	0.506 s	**74.09% faster**
DATA7 K=8192 peak RSS	1149.89 MiB	637.22 MiB	**44.58% lower**
DATA7 persistent state	512 MiB	64 KiB	**8192x smaller**

The progressive four-rung coverage path is exact but measures 0.709 s versus 0.643 s for repeated legacy scoring on this host, a 10.29% slowdown. No coverage-speed claim is made.

5 Compatibility and limits

PERF-P1 is execution-only Authority Class E. No existing scientific schema is versioned, and existing content digests remain authoritative. Approximate FPS, approximate neighbor queries, altered tie tolerance, altered coverage policy, and changed Stage-A semantics remain forbidden.

No authorizing MACE-MH-1 checkpoint or GPU runtime was supplied. This release makes no TRAIN2/EVAL2, GPU utilization, GPU memory, OOM, or complete production campaign performance claim.

6 Evidence

- Specification: `docs/specs/training_data/mlff_perf_p1_shared_exact_selection_spec.{md,pdf}`.
- Architecture: `docs/arch_manuals/mlff_training_data_architecture.{md,pdf}`.
- Revision note: `ARCHITECTURE_NOTES_MLFF_REV46.{md,pdf}`.
- CPU evidence: `audits/analysis/mlff_perf_p1_lta_cloud_cpu_2026-08-15.{json,md,pdf}`.
- Qualification: `release/MLFF_PERF_P1_QUALIFICATION_0.20.180a0.json`.

7 References

[1] T. F. Gonzalez, "Clustering to Minimize the Maximum Intercluster Distance," *Theoretical Computer Science* **38**, 293–306 (1985). DOI: 10.1016/0304-3975(85)90224-5.

[2] SciPy developers, "scipy.spatial.cKDTree.query," SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.cKDTree.query.html (accessed 2026-08-15).

[3] SciPy developers, "scipy.stats.wasserstein_distance," SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wasserstein_distance.html (accessed 2026-08-15).